SMOOTHNESS DESCENDS ALONG A FLAT MORPHISM OF SCHEMES LOCALLY OF FINITE TYPE OVER A FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof names the reduction and the descent input explicitly and even parenthesizes the second use of the equivalence; nothing is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-flat-under-smooth is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Equivalence between smoothness and geometric regularity at a point over a field. Same chapter, Lemma lemma-geometrically-regular-smooth, invoked at proof lines 4766-4767 and again at 4770-4772. Supplies the translation in both directions.
- Geometric regularity descends along a flat morphism. Same chapter, Lemma lemma-flat-under-geometrically-regular, invoked at proof lines 4768-4770. Supplies the descent step.

Source reference. Original lemma 05AW: sources/stacks-varieties/pinned-original-source.tex, source lines 2511–2511. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038X: sources/stacks-varieties/pinned-original-source.tex, source lines 2544–2544. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a morphism of schemes locally of finite type over k. Let $x \in X$ be a point and set $y = f(x)$. If $X \to \mathrm{Spec}(k)$ is smooth at x and f is flat at x then $Y \to \mathrm{Spec}(k)$ is smooth at y. In particular, if X is smooth over k and f is flat and surjective, then Y is smooth over k.*

Proof. It suffices to show that Y is geometrically regular at y, see Lemma 038X. This follows from Lemma 05AW (and Lemma 038X applied to (X, x)). □

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